FiGLi: Finsler Gauge on Lie algebra for Guiding Vector Fields

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***Abstract*—Guiding vector fields (GVFs) certify convergence to and traversal of a curve, but the certificate is usually tied to a fixed, hand-chosen distance. We show how to graft *learned* and *generative* components onto a constructive GVF on the special Euclidean group while preserving its convergence-and-traversal certificate exactly. The underlying theorem consumes the end-effector distance only through three structural axioms (left-invariance, chainability, local linearity), and on the one-parameter-subgroup connecting path chainability is *equivalent*, under mild regularity, to positive homogeneity of degree one. We therefore parameterize the distance by a homogeneous, positive-definite gauge (a Finsler norm) on the Lie algebra; such a gauge satisfies the three axioms for every parameter value, so the closed-loop field inherits the full certificate by construction rather than by post-hoc verification. We instantiate Mahalanobis and bounded radial gauges, prove that the admissible gauges are exactly the Finsler norms, and show the normal modes used for numerical stability only re-scale a positive gain. We lift the certificate to multi-robot formations on the product group, to slowly time-varying curves, and to bounded-error nearest-point search, and let the target curve be produced by a Riemannian flow-matching model. Throughout, the homogeneous gauges keep the chainability residual at machine precision ($\sim 10^{-15}$) for arbitrary parameters, whereas a non-homogeneous normalizing-flow distance trained to imitate the same metric remains $O(1)$ away – the quantitative meaning of an architectural certificate.**

The implementation is publicly available at: FiGLi

***Index Terms*—guiding vector fields, path following, Lie groups, certified learning, flow matching, Finsler gauges**

I. Introduction

Path following on the special Euclidean group $\mathrm{SE}(3)$ – driving a rigid body to a geometric curve of poses and then traversing it – is central to aerial, underwater, and space robotics and to manipulator end-effector control. Guiding vector fields (GVFs) are an attractive solution because they unify planning, following, and control in a single feedback law and decouple the spatial task from any time parametrization [1]–[4]. A recent constructive theory extends GVFs from Euclidean space to general matrix Lie groups and proves a convergence-and-traversal certificate for fully-actuated kinematic systems [5].

In parallel, learning and generative models have become powerful tools for specifying robot motion: stable vector fields can be learned on Lie groups [6], and flow-matching models generate distributions over $\mathrm{SE}(3)$ poses and trajectories [7]–[9]. The natural question is whether one can *learn* the geometry of a GVF – the distance that shapes convergence, or even the target curve – *without* sacrificing the formal guarantee. Penalizing a learned object to be "approximately" certified leaves an uncertified residual gap; what is needed is a guarantee that holds *by construction*.

This paper develops such a design under a single principle we call *learn inside the proof*: identify exactly which properties of the distance the certificate consumes, and constrain the learned object to satisfy those properties architecturally. The certificate of [5] uses the end-effector (EE) distance only through three axioms. We show (Sec. III) that, on the canonical connecting path, the decisive axiom – chainability – is equivalent under mild regularity to *positive homogeneity of degree one*, so any positive-definite homogeneous *gauge* on the Lie algebra induces an admissible distance for *every* parameter value. The contributions are:

- a transfer proposition showing that homogeneous, positive-definite gauges inherit the full certificate of [5], with an exact characterization of the admissible gauges as the (possibly asymmetric) Finsler norms;
- two architectural gauge families (a Mahalanobis norm and a bounded radial/Finsler gauge with a no-spurious-zero guarantee), and an analysis of the smoothing "normal modes" showing they only reshape a strictly positive gain;
- certified extensions to multi-robot formations on $\mathrm{SE}(3)^k$, slowly time-varying curves with feedforward, and bounded-error nearest-point search;
- the use of a Riemannian flow-matching model to *generate* the target curve, with the certificate reduced to a purely geometric, post-hoc properness check;
- numerical evidence that the homogeneous gauges sit at machine precision for arbitrary parameters while a non-homogeneous baseline is $O(1)$ away.

II. Related Work

Guiding vector fields. GVFs originate in Euclidean space [1], were made robust to spurious equilibria through a curvilinear/parametric distance [2], and were analyzed for global convergence and singularity elimination, including self-intersecting paths, by Yao *et al.* [4]. Kapitanyuk *et al.* [3] give a complete 2-D analysis with local input-to-state stability.

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Bartelt *et al.* [5] generalize the constructive field to connected matrix Lie groups with a non-redundant control input and prove the dichotomy convergence/traversal theorem we build on. Our work keeps that theorem intact and makes the *distance* learnable.

Learned and generative fields. Urain *et al.* [6] learn globally stable vector fields on Lie groups via diffeomorphisms; we instead enforce the structural axioms architecturally, avoiding a learned Lyapunov function. Flow matching [7] and its Riemannian generalization [8] train continuous normalizing flows by regressing conditional velocity fields; FoldFlow [9] specializes this to $\mathrm{SE}(3)^N$. We use a minimal Riemannian flow-matching model as a *curve generator*. Most closely related, the Score-Induced Guiding Vector Field (SGVF) [10] uses score-based diffusion to build guiding fields directly from point-cloud data distributions, targeting unordered, multi-branch, and probabilistic paths and relating score vanishing to GVF singularities. Our aim is complementary and the contrast is sharp: the generative component only *proposes* the curve, while the downstream field retains the closed-form convergence-and-traversal certificate of [5].

Composition and scalability. Multi-robot formation control via Lyapunov/potential arguments on product configuration spaces is classical [11], and symmetry reduction for left-invariant multi-agent systems on Lie groups is well studied [12]; we contribute an explicit homogeneous joint gauge with an *operative* joint-Lyapunov decrease (App. B). Fast exact and bounded-recall nearest-pose search on SE(3) [13] motivates our certified approximate-nearest result (App. A). Stability of the kinematic–dynamic composition follows from standard input-to-state stability [14] (App. C).

III. Method

A. Setup and the inherited certificate

Let $G = \mathrm{SE}(3)$ with Lie algebra $\mathfrak{se}(3) \cong \mathbb{R}^6$ and hat map $\mathcal{S} : \mathbb{R}^6 \to \mathfrak{se}(3)$. The fully-actuated kinematic system is $\dot{\mathbf{H}} = \mathcal{S}(\boldsymbol{\xi})\mathbf{H}$, $\mathbf{H} \in G$, with world-frame twist input $\boldsymbol{\xi} \in \mathbb{R}^6$ [5]. For the target curve $\mathcal{C} = \{\mathbf{H}_d(s)\}$ the field is

$$\Psi(\mathbf{H}) = k_N(\mathbf{H})\,\boldsymbol{\xi}_N(\mathbf{H}) + k_T(\mathbf{H})\,\boldsymbol{\xi}_T(\mathbf{H}), \qquad (1)$$

with $\boldsymbol{\xi}_N = -\mathbf{L}_{\mathbf{V}}[\widehat{D}](\mathbf{H}, \mathbf{H}_d(s^*))^\top$ the convergent component, $\boldsymbol{\xi}_T = \boldsymbol{\xi}_d(s^*)$ the curve tangent at the nearest pose, and gains $k_T > 0$, $k_N = 0$ on $\mathcal{C}$ and $k_N > 0$ off it. Here $\mathbf{L}$ is the left-invariant directional-derivative operator and $\widehat{D}$ an EE-distance [5]. The certificate consumes $\widehat{D}$ only through three axioms, stated on the one-parameter-subgroup connecting path

$$\Phi(\sigma, \mathbf{V}, \mathbf{W}) = \mathbf{V}\exp\big(\sigma \log(\mathbf{V}^{-1}\mathbf{W})\big), \quad \sigma \in [0,1] : \qquad (2)$$

(A1) left-invariance $\widehat{D}(\mathbf{AV}, \mathbf{AW}) = \widehat{D}(\mathbf{V}, \mathbf{W})$; **(A2)** chainability $\widehat{D}(\mathbf{V}, \mathbf{W}) = \widehat{D}(\mathbf{V}, \Phi_\sigma) + \widehat{D}(\Phi_\sigma, \mathbf{W})$; **(A3)** local linearity $\lim_{\sigma\to 0^+} \sigma^{-1}\widehat{D}(\mathbf{V}, \Phi_\sigma) > 0$ for $\mathbf{V} \neq \mathbf{W}$.

Theorem 1 ([5], Thm. III.17): If $\widehat{D}$ satisfies *(A1)–(A3)* and $\mathbf{H}_d$ is proper ($\boldsymbol{\xi}_d(s) \neq \mathbf{0}\ \forall s$), then the closed loop (1): *(i)* converges to $\mathcal{C}$ or to a singular set $\mathcal{P}$; *(ii)* $\mathcal{P}$ is escapable in finite time; *(iii)* on $\mathcal{C}$ the state traverses the curve.

For exponential groups the canonical instance is $\widehat{D}(\mathbf{V}, \mathbf{W}) = \|\log(\mathbf{V}^{-1}\mathbf{W})\|_F$ [5]. We write $\xi(\mathbf{Z}) \in \mathbb{R}^6$ for the twist coordinates of $\log \mathbf{Z}$ with the rotational block scaled by $\sqrt{2}$, so that $\|\xi(\mathbf{Z})\|_2 = \|\log \mathbf{Z}\|_F$.

B. Learned distances that inherit the certificate

Replace the fixed Frobenius norm by a learned *gauge* $N_\theta : \mathbb{R}^6 \to \mathbb{R}_{\geq 0}$ acting on the feature, $D_\theta(\mathbf{V}, \mathbf{W}) = N_\theta(\xi(\mathbf{V}^{-1}\mathbf{W}))$. Call N_θ *homogeneous* (property **(H)**) if it is positive-definite, positively homogeneous of degree one, and C^1 on $\mathbb{R}^6 \setminus \{0\}$.

Proposition 1 (Axiom inheritance): If N_θ satisfies *(H)*, then D_θ satisfies *(A1)–(A3)* on the path (2) for every parameter θ, and the field (1) inherits Theorem 1. Neither symmetry nor the triangle inequality is required.

Proof sketch: Fix $\mathbf{V}, \mathbf{W}$, let $a = \xi(\mathbf{V}^{-1}\mathbf{W})$ and $x = \log(\mathbf{V}^{-1}\mathbf{W})$. By the principal-log identity $\log(\exp(rx)) = rx$ for $r \in [0,1]$ [5], and since exponents along the subgroup commute,

$$D_\theta(\mathbf{V}, \Phi_\sigma) = N_\theta(\sigma a), \quad D_\theta(\Phi_\sigma, \mathbf{W}) = N_\theta((1-\sigma)a). \qquad (3)$$

(A1) is immediate from $\xi((\mathbf{AV})^{-1}\mathbf{AW}) = \xi(\mathbf{V}^{-1}\mathbf{W})$. By (3), *(A2)* reads $N_\theta(\sigma a) + N_\theta((1-\sigma)a) = N_\theta(a)$. Substituting $a \mapsto \mu a$ and writing $u = \sigma\mu$, $v = (1-\sigma)\mu$ turns this into Cauchy's equation $\gamma(u) + \gamma(v) = \gamma(u+v)$ on $[0,\infty)$ for $\gamma(t) = N_\theta(ta)$; continuity (from C^1 off 0 and $N_\theta(0) = 0$) forces $\gamma(t) = t\,N_\theta(a)$, i.e. positive homogeneity – and conversely homogeneity gives *(A2)*. Finally *(A3)* is $\lim_{\sigma\to 0^+} N_\theta(\sigma a)/\sigma = N_\theta(a) > 0$ by homogeneity and positive-definiteness. Theorem 1 then applies. ■

Chainability is thus governed by a *scalar* functional equation *on rays*; it is insensitive to cross-ray structure. This is why a homogeneity *penalty* only shrinks a residual, whereas homogeneity *by construction* satisfies (A2) identically. The same argument characterizes admissibility completely.

Proposition 2 (Characterization): A continuous N, C^1 off the origin, induces a $D(\mathbf{V}, \mathbf{W}) = N(\xi(\mathbf{V}^{-1}\mathbf{W}))$ satisfying *(A1)–(A3)* along (2) *iff* N is positively homogeneous of degree one and positive-definite, i.e. iff N is a (possibly asymmetric) Finsler norm on $\mathfrak{se}(3)$.

C. Two architectural gauges

Mahalanobis gauge $N_\theta(x) = \|\mathbf{L}_\theta x\|_2$, $\mathbf{L}_\theta$ lower-triangular with positive diagonal (Cholesky): a convex, symmetric norm, homogeneous and positive-definite for every θ; identity init gives $N_\theta = \widehat{D}$.

Radial (Finsler) gauge $N_\theta(x) = \|x\|_2\, h_\theta(x/\|x\|_2)$ with $h_\theta(u) = \exp(\log h_{\max} \cdot \tanh(\mathrm{MLP}_\theta(u))) \in [h_{\max}^{-1}, h_{\max}]$: positively homogeneous by inspection. The lower bound $h_\theta \geq h_{\max}^{-1}$ yields a *no-spurious-zero* guarantee,

$$N_\theta(x) \geq \|x\|_2/h_{\max} > 0 \quad \forall x \neq 0, \qquad (4)$$

so the learned gauge cannot create a false zero level set, independent of the network weights. Computing $x/\|x\|_2$ without an additive ε keeps homogeneity exact.

A generic invertible map (e.g. a RealNVP coupling flow) with $N(x) = \|\phi_\theta(x) - \phi_\theta(0)\|$ is left-invariant and positive-definite but *not* homogeneous; (A2) and (A3) fail globally, leaving only a local/tube guarantee. Section IV quantifies this gap.

D. Normal modes: direction invariance

For numerical smoothness the implemented normal component differentiates a scalar reshaping $g \circ D_\theta$ with $g \in \{g_{\text{true}}(d)=d,\ g_{\text{sm}}(d)=\sqrt{d^2+\varepsilon^2}-\varepsilon,\ g_{\text{sq}}(d)=d^2\}$.

Proposition 3 (Direction invariance / admissible regaining): Where $D_\theta > 0$ and differentiable, $\boldsymbol{\xi}_N = -g'(D_\theta)\,\mathbf{L}_{\mathbf{V}}[D_\theta]^\top = g'(D_\theta)\,\boldsymbol{\xi}_N^{\text{base}}$, where $\boldsymbol{\xi}_N^{\text{base}} = -\mathbf{L}_{\mathbf{V}}[D_\theta]^\top$ and $g' > 0$ on $(0,\infty)$. Hence the convergent *direction* is $\boldsymbol{\xi}_N^{\text{base}}$ for all three modes, orthogonality $\boldsymbol{\xi}_N \perp \boldsymbol{\xi}_T$ holds, and the closed loop equals the base field with normal gain $k_N(d) \mapsto g'(d)k_N(d)$. With the EC-distance $D_\theta(\mathbf{H}) = \min_s D_\theta(\mathbf{H}, \mathbf{H}_d(s))$ as Lyapunov function, its derivative is

$$\dot{D}_\theta = -\,k_N\, g'(D_\theta)\,\|\boldsymbol{\xi}_N^{\text{base}}\|^2 \le 0, \tag{5}$$

strictly negative off $\mathcal{C} \cup \mathcal{P}$.

The multipliers are $g'_{\text{true}} = 1$, $g'_{\text{sm}}(d) = d/\sqrt{d^2+\varepsilon^2} \in (0,1) \to 1$ for $d \gg \varepsilon$, $g'_{\text{sq}}(d) = 2d$. The smoothed mode (default) is C^∞ at $\mathcal{C}$ and matches the base field outside an $O(\varepsilon)$ tube; the squared mode is admissible but a genuinely different gain.

Exact gradient. Differentiating $g(N_\theta(\xi(\exp(\mathcal{S}(\delta))\mathbf{V})^{-1}\mathbf{W})))$ in δ at $\delta = 0$ through the closed-form $\mathrm{SE}(3)$ log yields the *exact* left-invariant gradient (the $\varepsilon_{\text{fd}} \to 0$ limit of a central difference, which is $O(\varepsilon_{\text{fd}}^2)$ accurate). This makes (5) hold for the *implemented* field. A safe-denominator (double-`where`) masking of the singular log branch is required so the gradient is finite at zero rotation.

E. Certified extensions

Multi-robot formations on $\mathrm{SE}(3)^k$**.** For k agents with stacked algebra $\mathbb{R}^{6k}$, use the Euclidean joint gauge $N_{\text{prod}}(x_1,\dots,x_k) = \big(\sum_i N_i(x_i)^2\big)^{1/2}$.

Lemma 1: If each N_i satisfies *(H)*, then N_{prod} satisfies *(H)* on $\mathbb{R}^{6k}$; in particular it is differentiable off the joint origin, unlike the literal sum $\sum_i N_i$, which has a gradient kink at partial-coincidence configurations.

By Proposition 1 applied on the product subgroup path, the certificate lifts to $\mathrm{SE}(3)^k$, provided the family of sub-curves is jointly proper. Moreover, with the joint Lyapunov function $V_{\text{prod}} = (\sum_i D_i^2)^{1/2}$, the stacked normal twist $(-\nabla D_i)_i$ obeys $\langle(-\nabla D_i)_i, -\nabla V_{\text{prod}}\rangle = \sum_i (D_i/V_{\text{prod}})\|\nabla D_i\|^2 > 0$ off the joint curve – a strict descent direction (an *operative* certificate, beyond axiom residuals; proof in App. B; cf. [11], [12]).

Slowly time-varying curves. For a moving target $\mathbf{H}_d(s,t)$ the frozen-t distance is admissible (Prop. 1), so with $V(\mathbf{H},t) = D_t(\mathbf{H})$,

$$\dot{V} = \underbrace{-k_N g'(D_t)\|\boldsymbol{\xi}_N^{\text{base}}\|^2}_{<0} + \partial_t V, \quad |\partial_t V| \le L_D \sup_s \|\partial_t \mathbf{H}_d\|, \tag{6}$$

with L_D a local Lipschitz constant of D_θ (finite by (H)). When the frozen decrease dominates the drift, a comparison argument [14] gives ultimate boundedness in a tube of radius $\delta = O(\text{path speed/rate})$, with traversal inside the tube. A feedforward $u_{\text{ff}} = (\partial_t \mathbf{H}_d)\mathbf{H}_d^{-1}$ cancels the leading drift. The constants in δ are not tightly identified; the certified content is qualitative (bounded tracking, error growing with drift speed, feedforward improvement).

Approximate nearest-pose search. If a fast oracle returns s_κ with $D_\theta(\mathbf{H}, \mathbf{H}_d(s_\kappa)) \le D_\theta(\mathbf{H}, \mathbf{H}_d(s^*)) + \kappa$, then by Lipschitz continuity of ∇D_θ away from the cut locus the normal twist obeys $\angle(\boldsymbol{\xi}_N^\kappa, \boldsymbol{\xi}_N^*) = O(\kappa/\mathrm{dist}(\mathbf{H}, \mathcal{C}))$. Since (5) is strictly negative, a small enough angular perturbation preserves descent outside a κ-enlarged tube: the loop still converges and traverses, and the steady true error grows $\sim$ linearly with κ. This certifies warm-started / bounded-recall search; the oracle may be any exact or approximate $\mathrm{SE}(3)$ search, e.g. [13] (proof in App. A).

Generative curves. The target $\mathbf{H}_d$ may be sampled from a Riemannian flow-matching model on $\mathrm{SE}(3)$ (an $\mathrm{SO}(3)$-geodesic + $\mathbb{R}^3$-linear interpolation, in the style of [8], [9]) and then resampled to a constant-speed parametrization. Because $\widehat{D}$ and the curve tangent are untouched, Theorem 1 (or Prop. 1) holds verbatim, conditional only on the realized curve being proper and free of $\mathrm{SE}(3)$ self-intersections – a purely geometric, post-hoc check, independent of the learner.

IV. Experiments

We reproduce the constructive machinery exactly and enter every learned object only as a black-box replacement of $\widehat{D}$. Unless noted, gauges use the smoothed mode ($\varepsilon = 10^{-6}$) and the exact gradient; the analytic Frobenius field is the reference overlay.

A. Axiom verification (the certificate gap)

Table I reports the maximum axiom residuals over random features/parameters. The homogeneous gauges (Mahalanobis, radial), at identity *and* random *and* task-trained parameters, keep the homogeneity and chainability residuals at machine precision. A RealNVP distance – even after being trained to imitate $\widehat{D}$ to a mean error of 2.5×10^{-3} – has homogeneity residual 1.85 and chainability residual 7.1×10^{-2}, i.e. $O(1)$. Fig. 1 visualizes the gap. Notably, ray-monotonicity passes at 100% *even for RealNVP*, so chainability, not monotonicity, is the discriminating test. The identity-initialized gauges reproduce $\widehat{D}$ to $\le 2 \times 10^{-13}$ and the analytic nearest pose exactly.

B. Normal modes and a task-shaped gauge

The `true` and `smoothed` modes reproduce the analytic steady-state distance (4.9×10^{-3} vs. baseline 5.1×10^{-3}), while

TABLE I
MAXIMUM AXIOM RESIDUALS (LOWER IS BETTER).

Gauge	homogeneity	chainability	cert.
Mahalanobis (identity)	1.8×10^{-15}	1.8×10^{-15}	full
Radial (identity)	1.8×10^{-15}	1.8×10^{-15}	full
Mahalanobis (random)	3.6×10^{-15}	1.8×10^{-15}	full
Radial (random)	1.1×10^{-14}	1.1×10^{-14}	full
Mahalanobis (task)	1.8×10^{-15}	1.8×10^{-15}	full
RealNVP (random)	1.7×10^{1}	2.4×10^{0}	local
RealNVP (trained)	1.9×10^{0}	7.1×10^{-2}	local

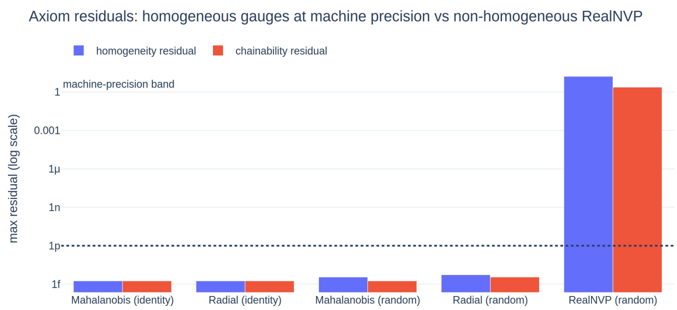

Fig. 1. Homogeneous gauges sit in the machine-precision band for arbitrary parameters; the non-homogeneous RealNVP distance is $O(1)$ away. The certificate is architectural, not statistical.

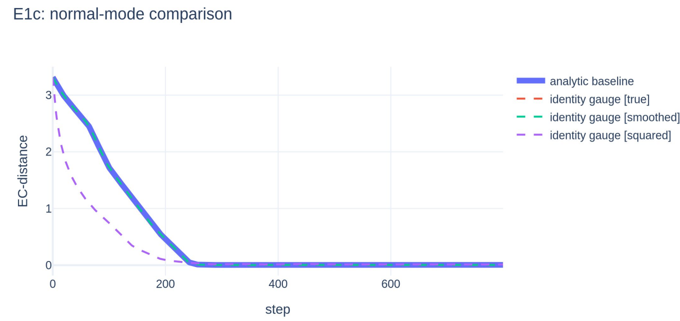

Fig. 2. Normal-mode comparison (identity gauge). The `true` and `smoothed` modes overlay the analytic baseline; `squared` converges faster early (its $2d$ multiplier) but to a different gain, all sharing the same convergent direction (Prop. 3).

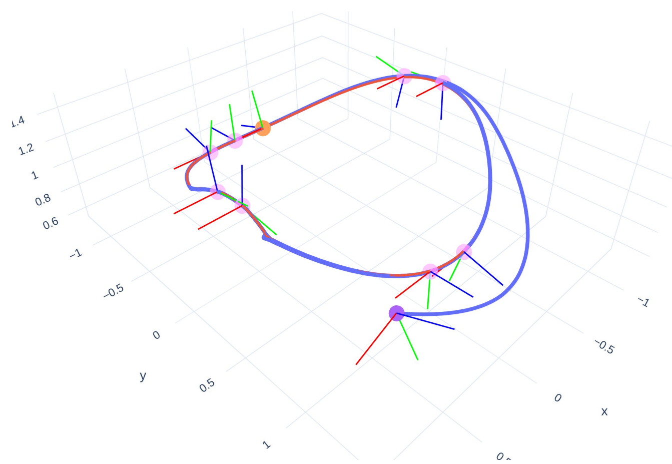

Fig. 3. Tracking a *generated* $\mathrm{SE}(3)$ curve with the unchanged certified field (frames: red/green/blue body axes). The certificate holds because the realized curve is proper.

`squared` settles higher (1.3×10^{-2}), exactly as predicted by the $2d$ multiplier of Prop. 3; gradient-direction cosines equal 1.000000. Fitting a Mahalanobis gauge to a non-Frobenius anisotropic target (weighting height and one rotation axis) reaches a mean error of 6.3×10^{-4}, recovers the target diagonal $(1,1,\sqrt{3},1,1,1.58)$, changes the approach geometry (cosine 0.95 to the isotropic field), *and* keeps the residuals of Table I unchanged while still converging and traversing.

C. Generative curve (D2)

A flow-matching model trained on a family of curves generates a coherent $\mathrm{SE}(3)$ curve; a dedupe-plus-arc-length fitter yields a proper parametrization ($\min\|\boldsymbol{\xi}_d\| = 0.46 > 0$, arc-length CoV ≈ 0). Fed into the unchanged analytic field, the system converges to 4.3×10^{-3} and completes multiple laps (Fig. 3); three independently sampled curves are all proper and converge.

D. Ablations (D3, D4)

The RealNVP distance (D3) converges *near* the curve but carries only the local guarantee implied by its $O(1)$ residuals. A field learned *directly* (D4) to imitate Ψ has no off-support Lyapunov guarantee; from out-of-distribution initial poses the certified gauge converges to a markedly smaller steady error and a higher lap count than the direct field (Fig. 4).

E. Scalability and certified approximate nearest

An exact GPU-batched nearest-pose query is bit-identical to the brute-force search and up to $\sim 290\times$ faster at 8000 samples, while a warm-started local search is essentially curve-length-independent. Replacing the exact search by a κ-approximate oracle preserves convergence and traversal for all tested $\kappa \in [0, 0.4]$, with the steady *true* error growing sub-linearly below the reference slope κ (Fig. 5), confirming the graceful-degradation result of Sec. III-E.

F. Multi-robot and time-varying

For a two-robot formation, the joint gauge has homogeneity/chainability residuals at machine precision ($\sim 10^{-15}$), both agents converge and traverse (~ 11 laps), the joint distance is bounded away from zero over random joint poses, and the joint normal twist is a strict descent direction of V_{prod} (minimum cosine $0.70 > 0$). For a drifting helix the steady error grows roughly linearly with drift speed (slope 0.28, $R = 0.95$) and the feedforward term reduces it at slow speeds (Fig. 6), matching the perturbation analysis of (6).

V. DISCUSSION

The results delineate cleanly what is unconditional from what is not. *Unconditional (architecture):* for the homogeneous gauges, the three axioms hold to machine precision for all parameters, and with the exact gradient the implemented

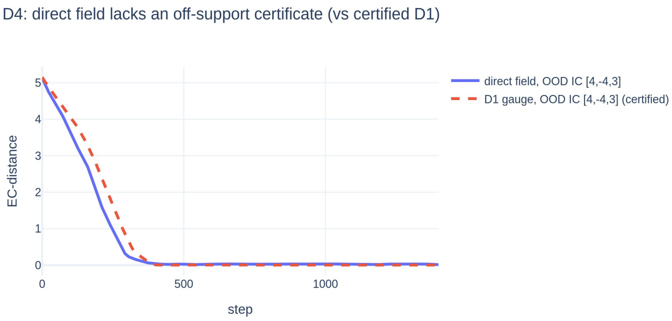

Fig. 4. From an out-of-distribution initial pose, the certified D1 gauge retains convergence and traversal; the directly-learned field lacks an off-support certificate.

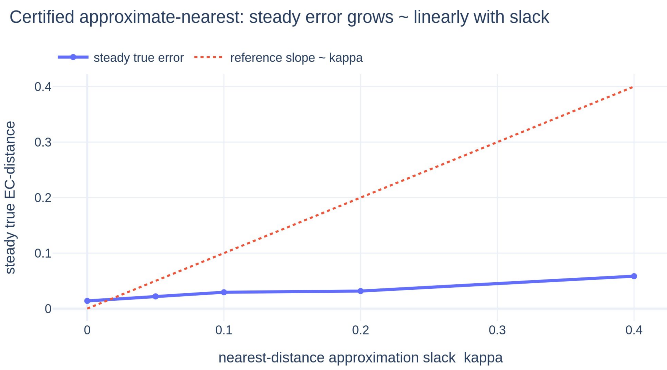

Fig. 5. Certified approximate-nearest: convergence and traversal persist; the steady true error grows roughly linearly with the approximation slack κ.

convergent direction equals $-\nabla(g \circ D_\theta)$ exactly, so the strict Lyapunov decrease (5) holds for the implemented field; Prop. 3 governs the modes. *Conditional (geometry):* the generative-curve guarantee requires the realized curve to be proper with no $\mathrm{SE}(3)$ self-intersection – a post-hoc, gauge-independent check that an under-trained generator can fail. *Approximate (honest):* the time-varying and approximate-nearest results are qualitative; their constants are not identified, and the experiments report the predicted trends rather than certified bounds. *Out of scope:* dynamics, actuation limits, obstacles, sensing (the field is a first-order kinematic law), and the unavoidable antipodal ambiguity of the $\mathrm{SE}(3)$ log at exactly $\theta = \pi$.

A limitation of the framing is that "learning" here shapes geometry (the gauge) or proposes the curve, rather than learning the control law end-to-end; this is deliberate, and is precisely what buys the architectural certificate. The composition corollaries (a learned visual gauge in front of N_θ; an ISS dynamic layer behind the kinematic field [14]) indicate how the guarantee can be carried into richer pipelines without a learned Lyapunov function (formalized in App. C).

VI. Conclusion

We showed that the convergence-and-traversal certificate of a constructive SE(3) guiding vector field transfers, by architecture, to learned and generative components, once the distance

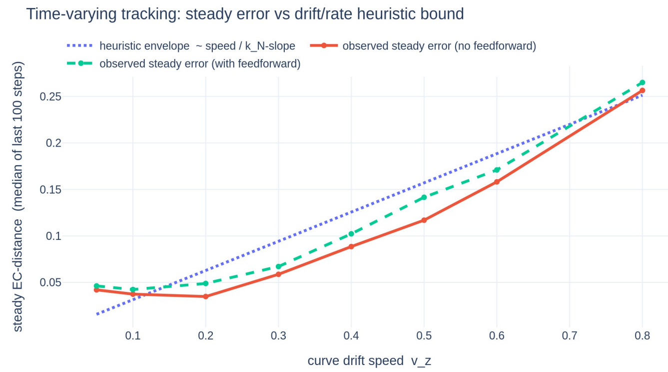

Fig. 6. Time-varying tracking: the steady error tracks a drift/rate heuristic in slope and magnitude; feedforward tightens it. The bound's constants are not tightly identified.

is realized as a positive-definite homogeneous (Finsler) gauge on the Lie algebra. The decisive step is the equivalence, on the connecting path, between chainability and positive homogeneity of degree one. The construction lifts to multi-robot formations, slowly time-varying curves, and bounded-error nearest-pose search, and accommodates a flow-matching curve generator. Numerically, the homogeneous gauges remain certified at machine precision for arbitrary parameters, while a non-homogeneous learned distance does not – the quantitative signature of an architectural, rather than statistical, guarantee. Future work includes tight constants for the time-varying and approximate-nearest bounds, and certified end-to-end visual gauges.

Appendix

The following formalizes the extension claims of Sec. III-E; the body states them compactly and the proofs are collected here. Numbering continues from the main propositions. Throughout, $D_\theta(\mathbf{H}) = \min_s D_\theta(\mathbf{H}, \mathbf{H}_d(s))$ is the EC-distance of an admissible gauge (property (H)) and the curve is proper.

A. Certified approximate-nearest search

Proposition 4 (Graceful degradation): Suppose a fast oracle returns s_κ with $D_\theta(\mathbf{H}, \mathbf{H}_d(s_\kappa)) \le D_\theta(\mathbf{H}) + \kappa$, and let $\boldsymbol{\xi}_N^\kappa = -\mathbf{L}_{\mathbf{V}}[D_\theta](\mathbf{H}, \mathbf{H}_d(s_\kappa))^\top$ be the resulting normal twist (versus the exact $\boldsymbol{\xi}_N^*$ at s^*). Then there exist $\kappa_0 > 0$ and a class-$\mathcal{K}$ function $\delta(\kappa)$ with $\delta(\kappa) \to 0$ as $\kappa \to 0$ such that, for all $\kappa < \kappa_0$, the closed loop converges to the $\delta(\kappa)$-tube about $\mathcal{C}$, escapes the (enlarged) singular set in finite time, and traverses inside the tube; the steady true distance is $O(\kappa)$.

Proof: (i) Smoothness off the curve. By (H), N_θ is positively homogeneous of degree one and C^1 off the origin, so $\mathbf{H} \mapsto \nabla_{\mathbf{H}} D_\theta(\mathbf{H})$ is locally Lipschitz on compact sets bounded away from the cut locus, in particular where $D_\theta \ge \delta_0 > 0$. *(ii) Angular bound.* A first-order expansion of ∇D_θ about the true nearest pose gives $\angle(\boldsymbol{\xi}_N^\kappa, \boldsymbol{\xi}_N^*) = O\big(\kappa/\mathrm{dist}(\mathbf{H}, \mathcal{C})\big)$, the hidden constant being the local Lipschitz constant of ∇D_θ. *(iii) Perturbed decrease.* By Prop. 3, off $\mathcal{C} \cup \mathcal{P}$ the nominal derivative satisfies $\dot{V} = -k_N \|\boldsymbol{\xi}_N^*\|^2 \le -c\, k_N \|\boldsymbol{\xi}_N^*\|^2 < 0$; the perturbed field replaces $\boldsymbol{\xi}_N^*$ by $\boldsymbol{\xi}_N^\kappa$, and

$$\left|\langle \boldsymbol{\xi}_N^\kappa, \boldsymbol{\xi}_N^* \rangle - \|\boldsymbol{\xi}_N^*\|^2\right| \le \|\boldsymbol{\xi}_N^*\| \cdot O(\kappa/D_\theta) \cdot \|\boldsymbol{\xi}_N^*\|,$$

so for any fixed gain schedule there is $\kappa_0 > 0$ (depending on the local Lipschitz constants and $\min k_N$ outside a compact set) for which the perturbed derivative stays strictly negative whenever $D_\theta \ge \delta(\kappa)$, with $\delta(\kappa) \to 0$. *(iv) Ultimate boundedness and traversal.* The comparison lemma [14, Ch. 9] yields entry into and invariance of a $\delta(\kappa)$-tube about $\mathcal{C}$; inside the tube the traversal property holds up to an arbitrarily small lag by continuous dependence on the field, and finite-time escape from the slightly enlarged singular set follows from the same perturbation argument as [3]. ■

The slope $\delta(\kappa) \sim \kappa$ matches the measured steady-error growth (Fig. 5; v9 also reports an analogous slope ≈ 0.28 for the time-varying case).

B. Multi-robot operative certificate

Proof of Lemma 1: Write $N_{\mathrm{prod}}(X) = \big(\sum_{i=1}^k N_i(X_i)^2\big)^{1/2}$ for $X = (X_1, \ldots, X_k) \in \mathbb{R}^{6k}$. Positive homogeneity of degree one and positive-definiteness are inherited componentwise from each N_i. Each N_i^2 is homogeneous of degree two and C^1 on all of $\mathbb{R}^6$ (its gradient $2N_i \nabla N_i \to 0$ at the origin), and $t \mapsto \sqrt{t}$ is smooth for $t > 0$; hence N_{prod} is C^1 wherever $\sum_i N_i^2 > 0$, i.e. off the joint origin. By contrast the literal sum $\sum_i N_i$ is non-differentiable at every *partial*-coincidence configuration (any single $X_i = 0$), where the conic non-smoothness of that N_i produces a one-sided gradient jump. ■

On the product one-parameter-subgroup path the stacked feature scales as σX, so homogeneity of N_{prod} yields chainability and local linearity exactly as in Prop. 1 (left-invariance is componentwise); Theorem 1 therefore lifts to $\mathrm{SE}(3)^k$ under joint properness of the sub-curve family.

Proposition 5 (Operative joint descent): Let $\boldsymbol{\xi}_N^i = -\nabla_{\mathbf{H}_i} D_i$ be the descent twist of agent i and $V_{\mathrm{prod}} = (\sum_i D_i^2)^{1/2}$. Then off the joint curve ($V_{\mathrm{prod}} > 0$),

$$\big\langle (\boldsymbol{\xi}_N^i)_i,\, -\nabla V_{\mathrm{prod}} \big\rangle = \sum_{i=1}^k \frac{D_i}{V_{\mathrm{prod}}} \, \|\boldsymbol{\xi}_N^i\|^2 > 0,$$

i.e. the implemented stacked normal field is a strict descent direction of the joint Lyapunov function, not merely axiom-compliant.

Proof: $\nabla_{\mathbf{H}_i} V_{\mathrm{prod}} = (D_i / V_{\mathrm{prod}}) \nabla_{\mathbf{H}_i} D_i$, so $-\nabla V_{\mathrm{prod}} = \big(-(D_i/V_{\mathrm{prod}}) \nabla D_i\big)_i$. Pairing with $(\boldsymbol{\xi}_N^i)_i = (-\nabla D_i)_i$ gives the stated sum; each term is nonnegative and at least one is positive when $V_{\mathrm{prod}} > 0$. ■

v9 measures the minimum of this normalized inner product over 200 random off-curve joint poses and obtains $0.70 > 0$, an empirical operative certificate.

C. Composition: visual gauge and ISS dynamics

Corollary 1 (Front/back composition): (i) Let $\phi_{\mathrm{vis}} : \mathcal{I} \to \mathbb{R}^6$ be any (learned) feature map. If the fusion $N_\theta \circ \phi_{\mathrm{vis}}$ is positively homogeneous of degree one and positive-definite, then the composite distance satisfies *(A1)–(A3)* and the field inherits Theorem 1. *(ii)* If the kinematic field produces a reference velocity $v_{\mathrm{ref}}(\mathbf{H})$ and the actuated dynamics are ISS with respect to the velocity-tracking error with gain γ, then the cascaded kinematic–dynamic system is ultimately bounded in a tube whose radius is governed by γ and the kinematic convergence rate.

Proof sketch: (i) is Prop. 1/2 applied to the composite gauge: only homogeneity and positive-definiteness of the final scalar are used, so any front-end map preserving them is admissible. (ii) is the standard ISS cascade [14, Ch. 4.9, Thm. 4.19]: the kinematic GVF is an ISS-stable outer loop generating v_{ref}, and the ISS inner loop tracks it with a bound proportional to γ; neither step requires a learned Lyapunov function, unlike end-to-end learned fields [6]. ■

Remark (tighter bounds). The tube radii in Prop. 4 and (6) can be made explicit from (a) the local Lipschitz constant of D_θ (finite by (H)), (b) the minimum slope of $k_N(d)$ near zero, and (c) a path-speed bound, via standard singular-perturbation/comparison analysis [14, Ch. 9, 11], [3]; we leave the constant identification to future work.

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